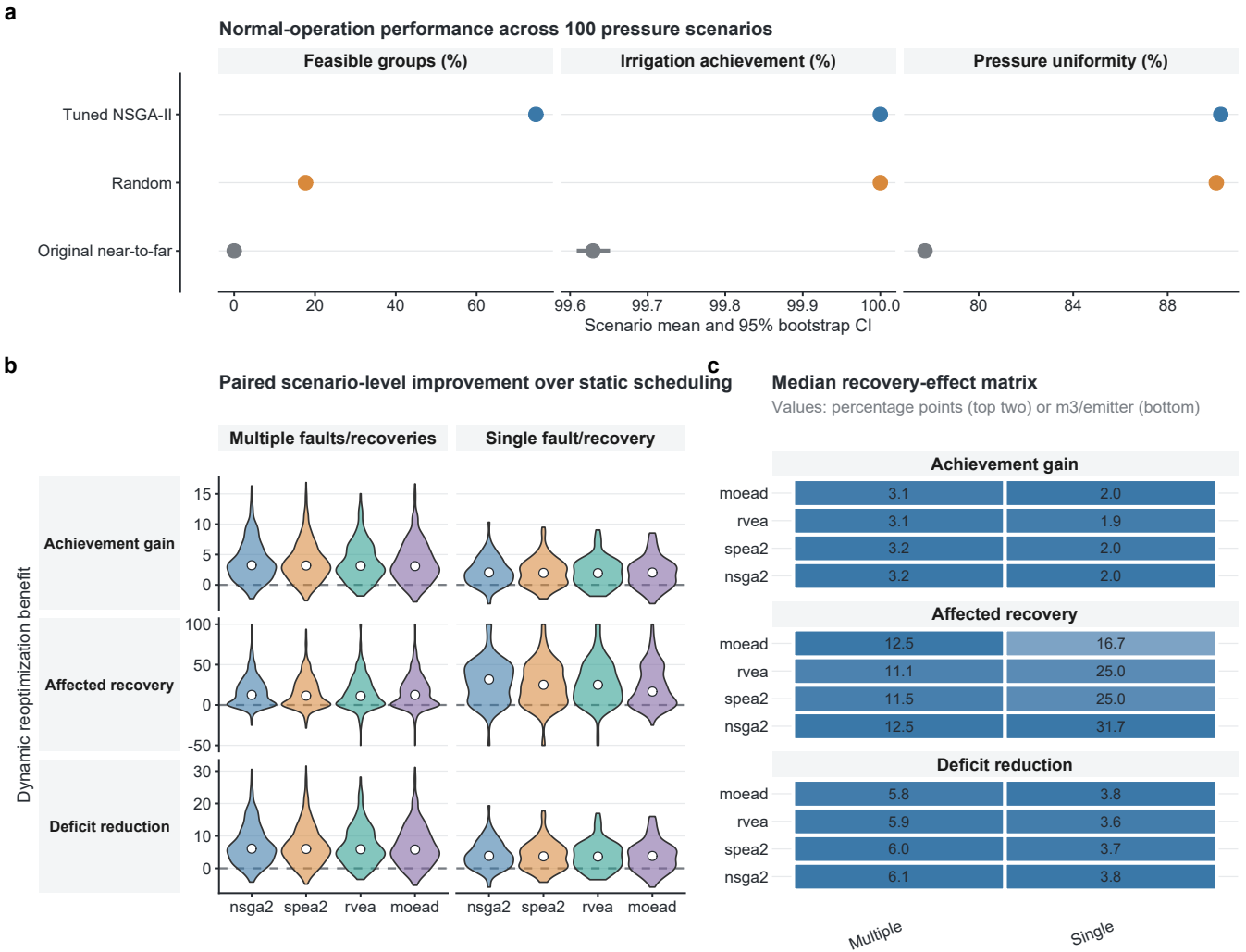

a
Normal-operation performance across 100 pressure scenarios
Feasible groups (%)
Irrigation achievement (%)
Pressure uniformity (%)
Tuned NSGA-II
Random
Original near-to-far
0
20
40
60
99.6
99.7
99.8
99.9
100.0
80
84
88
Scenario mean and 95% bootstrap CI
b
Paired scenario-level improvement over static scheduling
Multiple faults/recoveries
Single fault/recovery
Dynamic reoptimization benefit
Achievement gain
Affected recovery
Deficit reduction
15
10
5
0
100
50
0
-50
30
20
10
0
nsga2
spea2
rvea
moead
c
Median recovery-effect matrix
Values: percentage points (top two) or m3/emitter (bottom)
Achievement gain
moead 3.1 2.0
rvea 3.1 1.9
spea2 3.2 2.0
nsga2 3.2 2.0
Affected recovery
moead 12.5 16.7
rvea 11.1 25.0
spea2 11.5 25.0
nsga2 12.5 31.7
Deficit reduction
moead 5.8 3.8
rvea 5.9 3.6
spea2 6.0 3.7
nsga2 6.1 3.8
Multiple
Single